Gowin FIFO HS

User Guide

IPUG760-1.2E, 07/01/2021

Revision History

Date	Version	Description
01/15/2021	1.0E	Initial version published.
02/04/2021	1.1E	The description of Chapter 4 updated.
07/01/2021	1.2E	● FIFO SC HS IP timing figures in Chapter 4 updated. ● FIFO HS IP configuration interface in Chapter 5 added.

Contents

List of Figures

List of Tables

1 About This Guide

1.1 Purpose

The purpose of Gowin FIFO HS IP User Guide is to help you to quickly learn the function and usage of Gowin FIFO HS IP by providing descriptions of features, ports, timing, GUI and reference design.

1.2 Related Documents

The latest user guides are available on the GOWINSEMI Website. You can find the related documents at www.gowinsemi.com:

1. DS100, GW1N series of FPGA Products Data Sheet
2. DS117, GW1NR series of FPGA Products Data Sheet
3. DS102, GW2A series of FPGA Products Data Sheet
4. DS226, GW2AR series of FPGA Products Data Sheet

1.3 Terminology and Abbreviations

The terminology and abbreviations used in this manual are as shown in Table 1-1.

Table 1-1 Abbreviations and Terminology

Terminology and Abbreviations	Meaning
FIFO	First Input First Output
IP	Intellectual Property
RAM	Random Access Memory
LUT	Look-up Tables
GSR	Global System Reset
ECC	Error Correcting Code

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly by the following ways.

Website: www.gowinsemi.com/en

E-mail: support@gowinsemi.com

2 FIFO Overview

2.1 FIFO

FIFO is the abbreviation of First In First Out. It is a First In First Out sequence. Peripheral control logic is responsible for reading from or writing to FIFO. FIFO also provides many handshake signals that interact with the peripheral control logic.

When FIFO is not full and write-enable is active, data is written to the FIFO storage queue at the rising edge of the CLK. Full flag indicates that FIFO is full and no more write operations will be executed.

When FIFO is not empty and read-enable is active, data is read from the FIFO storage queue at the rising edge of the CLK. An empty flag indicates FIFO is empty and no more read operations will be executed.

Illegal requests will not affect FIFO. If a read request is initiated when the FIFO is empty or a write request is initiated when the FIFO is full, they will not have any influence on the data in FIFO. The operations will be ignored, and the overflow and underflow flags will be set. You can determine whether the operation is illegal or not by monitoring these two signals. The water level signal will indicate how much effective data FIFO directly contains. You can also customize empty or full flags by setting different thresholds.

FIFO HS contains both synchronous FIFO (FIFO SC HS IP) and asynchronous FIFO (FIFO HS IP). The structures of these are shown in Figure 2-1 and Figure 2-2 respectively.

Figure 2-1 FIFO HS IP Diagram

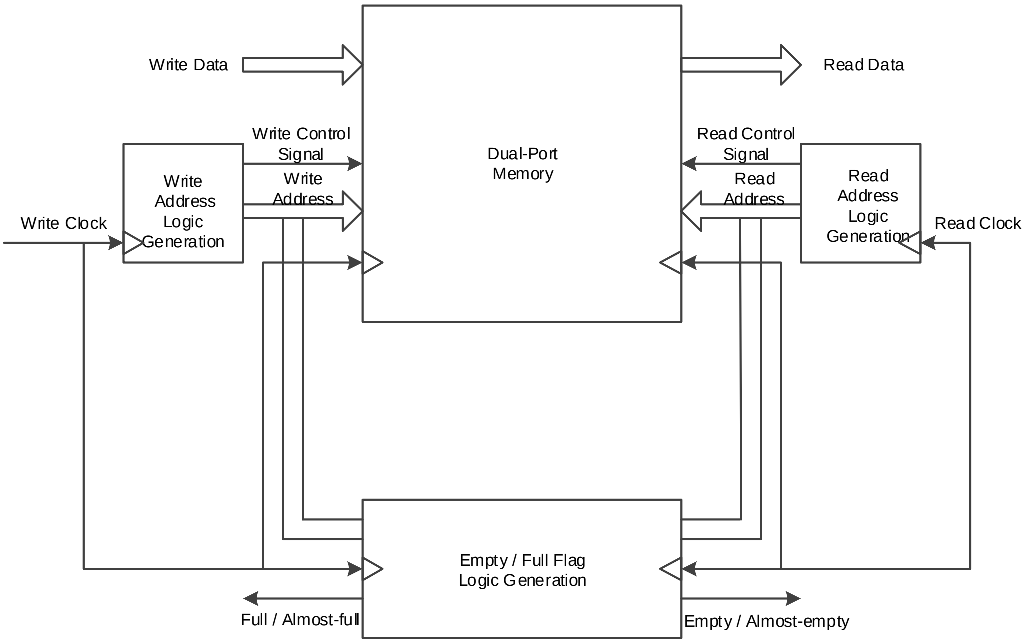
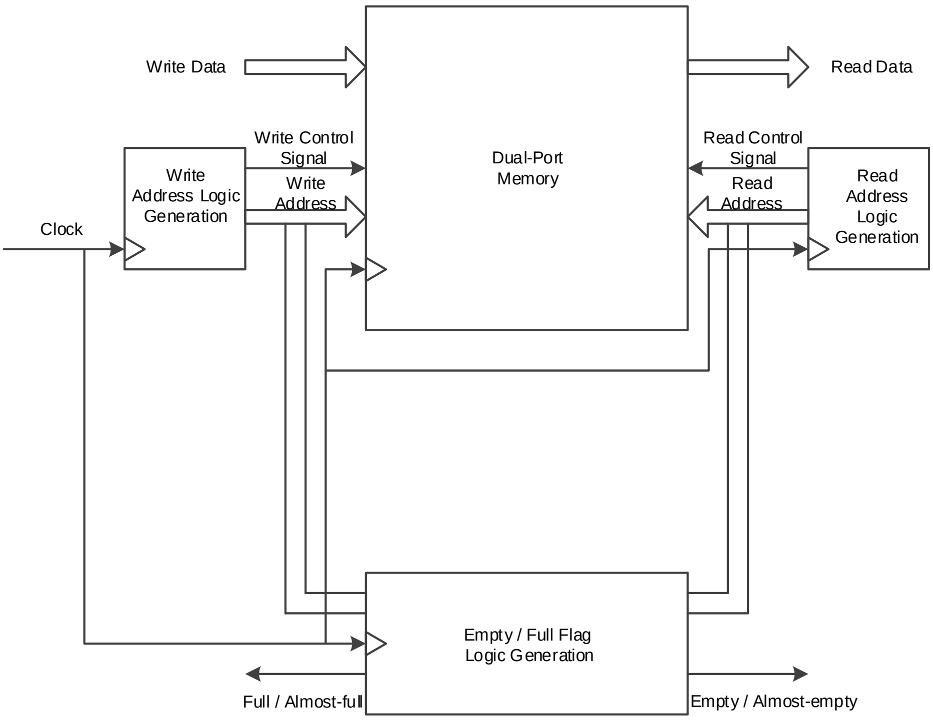


Figure 2-2 FIFO SC HS IP Diagram



2.2 FIFO HS/FIFO SC HS IP

Gowin FIFO IP contains synchronous FIFO (FIFO SC HS) IP and asynchronous FIFO (FIFO HS) IP.

- The read ports signals and write ports signals of the synchronous FIFO are all controlled by one clock domain;

- The read ports signals and write ports signals of the asynchronous FIFO are controlled by two independent clock domains.

The Gowin IP Core Generator can be used to generate a FIFO IP with single-cycle read/write and configurable ports. The configuration port is such that you can generate FIFO IP of different bit widths and data depths according to your needs.

Table 2-1 FIFO HS/FIFO SC HS IP Overview

FIFO HS IP and FIFO SC HS IP	
IP Core Application	
Logic Resource	Different resources for different configurations, different bit widths and data depths.
Delivered Doc.	-
Design Files	Verilog (encrypted)
Verilog	-
Synthesis Software	GowinSynthesis®
Application Software	Gowin Software (V1.9.8Beta)

2.3 FIFO HS/FIFO SC HS IP Function and Feature

2.3.1 FIFO HS IP Function and Feature

Gowin asynchronous FIFO HS IP can transfer and cache the data of different bit width in asynchronous clock domains and configure different output control signals and data structures according to your requirements.

The main features of the FIFO IP are as follows:

- The implementation type of the internal storage structure of FIFO HS IP is configurable, including block SRAM, Shadow SRAM and LUT;
- Write data depth is configurable. The depth is 2^n, and the maximum value is 65536;
- Write data bit width is configurable. The size is 1-256 bit;
- Read data depth is configurable. The depth is 2^n, and the maximum value is 65536;
- Read data bit-width=Write data depth x Write bit-width / Read data depth, not configurable.

  Note!

  The figures applied to the formula described above must be divisible, which limits the maximum read data depth.
- The output of the read/write data number is configurable. You can choose whether to output the read/write number data or not;
- The reset function is configurable. You can choose not to use reset (reset by GSR), to use one reset, or to reset read and write respectively;
- The flag signal output is optional. You can choose to output a almost-empty flag, an almost-full flag or no flag; The output almost-empty signal and almost-full signal are one cycle later compared to those of the original FIFO IP;
- If you choose to output an almost-empty or almost-full flag, the threshold of almost-empty, almost-full is configurable. They can be set as: Static single constant threshold, Static double constant threshold, Dynamic single input threshold, Dynamic double input threshold;
- The ECC check function is optional. The ECC check function can be selected when the asynchronous FIFO HS IP storage structure is achieved by block SRAM and the read/write bit-width are equal and are all below than 64bit;
- The output register function is configurable. If you choose the output

register function, the read-enable (RdEn) control is optional. If you choose enable-control, the output register is controlled by RdEn, and the last data cannot be output. If you choose the output register function instead of read-enable control, the read data output will be a cycle later, and the last data will be output.

- First-Word Fall-Through function is supported, but some conditions of this timing are not easily met and the design runs at a slower clock frequency.

2.3.2 FIFO SC HS IP Function and Feature

Gowin synchronous FIFO SC HS IP can transfer and cache the data of the same bit width and depth in the same clock domains. It can also configure and output different control signals according to your requirements.

The main features of the FIFO IP are as follows:

- The implementation type of the internal storage structure of FIFO SC HS IP is configurable, including block SRAM, Shadow SRAM and LUT;
- Write data depth is configurable. The depth is 2^n, and the maximum value is 65536;
- Write data bit width is configurable. The size is 1-256 bit;
- The depth and bit-width of read/write data are equal;
- The output of the write data number is configurable. Users can choose whether to output the write number data;
- The flag signal output is optional. You can choose to output an almost-empty flag, an almost-full flag or no flag;
- If you choose to output an almost-empty or almost-full flag, the threshold of almost-empty, almost-full is configurable. They can be set as: Static single constant threshold, Static double constant threshold, Dynamic single input threshold, Dynamic double input threshold;
- The ECC check function is optional. The ECC check function can be selected when the FIFO SC HS storage structure is achieved by block SRAM and the read/write bit-width are equal and are all below than 64bit;
- The output register function is configurable. If you choose the output register function, the read-enable (RdEn) control is optional. If the user chooses enable-control, the output register is controlled by RdEn, and the last data cannot be output. If you choose the output register function instead of read-enable control, the read data output will be a cycle later, and the last data will be output.

- First-Word Fall-Through function is supported.

2.4 The maximum Frequency and Resource Utilization of FIFO HS IP

Gowin FIFO is implemented by Verilog. The Max. frequency of FIFO HS IP changes according to the data depth, bit-width, and the speed level of the devices in use. Performance and resource utilization may vary when you use different devices of varying density, speed, or grade.

The resource utilization and performance of FIFO HS IP are shown in Table 2-2 and Table 2-3. This enhancement is only for the FIFO Type of BSRAM, so the FIFO Type of Block SRAM is used as an example.

Table 2-2 FIFO HS IP with No Optional Feature

FIFO Type	Depth x Width	FPGA Series	Performance (MHz)	Resource Utilization				
				Reg	LUT	ALU	Logics	BSRAM
Block SRAM	4096 x 16	GW2A-55-484	WrClk=124.243 RdClk=122.991	54	35	39	74	4
	512 x 16		WrClk=195.025 RdClk=186.386	42	27	30	57	1
	64 x 16		WrClk=117.189 RdClk=117.054	30	19	21	40	1
	4096 x 16	GW2A-18-484	WrClk=164.473 RdClk=147.242	54	35	39	74	4
	512 x 16		WrClk=201.602 RdClk=208.362	42	27	30	57	1
	64 x 16		WrClk=149.176 RdClk=171.152	30	19	21	40	1
	4096 x 16	GW1N-4-144	WrClk=144.579 RdClk=115.026	54	35	39	74	4
	512 x 16		WrClk=141.036 RdClk=143.781	42	27	30	57	1
	64 x 16		WrClk=134.581 RdClk=127.163	30	19	21	40	1

Table 2-3 FIFO HS IP with Optional Feature

FIFO Type	Depth x Width	FPGA Series	Performance (MHz)	Resource Utilization					Configuration		
				Reg	LUT	ALU	Logics	BSRAM	Data NUM	Flag con	ECC
BSRAM	4096 x 16	GW2A-55-484	WrClk=158.200 RdClk=201.117	162	141	82	223	4	Yes	Yes	Yes
	512 x 16		WrClk=249.642 RdClk=168.863	126	122	64	186	1	Yes	Yes	Yes
	64 x 16		WrClk=250.041 RdClk=176.390	90	110	46	156	1	Yes	Yes	Yes
	4096 x 16	GW2A-18-484	WrClk=202.941 RdClk=162.838	162	141	82	223	4	Yes	Yes	Yes
	512 x 16		WrClk=263.991 RdClk=170.131	126	122	64	186	1	Yes	Yes	Yes
	64 x 16		WrClk=253.679 RdClk=178.594	90	110	46	156	1	Yes	Yes	Yes
	4096 x 16	GW1N-4-144	WrClk=116.434 RdClk=83.14	162	141	82	223	4	Yes	Yes	Yes
	512 x 16		WrClk=124.109 RdClk=86.073	126	122	64	186	1	Yes	Yes	Yes
	64 x 16		WrClk=119.766 RdClk=82.506	90	110	46	156	1	Yes	Yes	Yes

Note!

Application verification of FIFO on other GOWIN FPGA devices will be released in succession.

2.5 The Maximum Frequency and Resource Utilization of FIFO SC HS IP

Gowin synchronous FIFO is implemented by Verilog. The Max. frequency of FIFO SC HS changes according to the data depth, bit-width, and the speed level of the devices in use. Performance and resource utilization may vary when you use different devices of varying density, speed, or grade.

The resource utilization and performance of FIFO SC HS are shown in Table 2-4 and Table 2-5. This optimization is only for the FIFO Type of BSRAM, so the FIFO Type of Block SRAM is used as an example.

Table 2-4 FIFO SC HS IP with No Optional Feature

FIFO Type	Depth x Width	FPGA Series	Performance (MHz)	Resource Utilization				
				Reg	LUT	ALU	Logics	BSRAM
Block SRAM	4096 x 16	GW2A-55-484	184.072	54	13	65	78	4
	512 x 16		187.569	42	11	50	61	1
	64 x 16		211.382	30	9	35	44	1
	4096 x 16	GW2A-18-484	193.129	54	13	65	78	4
	512 x 16		201.438	42	11	50	61	1
	64 x 16		236.317	30	9	35	44	1
	4096 x 16	GW1N-4-144	101.041	54	13	65	78	4
	512 x 16		115.146	42	11	50	61	1
	64 x 16		120.803	30	9	35	44	1

Table 2-5 FIFO SC HS IP with Optional Feature

FIFO Type	Depth x Width	FPGA Series	Performance (MHz)	Resource Utilization					Configuration		
				Reg	LUT	ALU	Logics	BSRAM	Data NUM	Flag con	ECC
BSRAM	4096 x 16	GW2A-55-484	136.188	71	110	83	193	4	Yes	Yes	Yes
	512 x 16		166.123	56	105	65	170	1	Yes	Yes	Yes
	64 x 16		144.873	41	102	47	149	1	Yes	Yes	Yes
	4096 x 16	GW2A-18-484	158.949	71	110	83	193	4	Yes	Yes	Yes
	512 x 16		172.282	56	105	65	170	1	Yes	Yes	Yes
	64 x 16		161.652	41	102	47	149	1	Yes	Yes	Yes
	4096 x 16	GW1N-4-144	104.066	69	19	78	97	4	Yes	Yes	Yes
	512 x 16		114.538	54	16	60	76	1	Yes	Yes	Yes
	64 x 16		105.768	39	12	42	54	1	Yes	Yes	Yes

Note!

Application verification of FIFO SC HS on other GOWIN FPGA devices will be released in succession.

3 Port Description

3.1 FIFO HS IP Port

A description of the FIFO HS IP I/O is presented in Table 3-1.

Table 3-1 FIFO HS IP IO Post

Signal	Data Width	I/O	Initial State	Optional	Description
Data	[WDSIZE-1:0]	I	-	No	Write data
WrClk	1	I	-	No	Write clock
RdClk	1	I	-	No	Read clock
WrEn	1	I	-	No	Write enable
RdEn	1	I	-	No	Read enable
Reset	1	I	-	Yes	Reset, active high
WrReset	1	I	-	Yes	Write reset, active high
RdReset	1	I	-	Yes	Read reset, active high
AlmostEmptySetTh	[RASIZE-1:0]	I	-	Yes	Almost-empty flag, set threshold to 1
AlmostEmptyClrTh	[RASIZE-1:0]	I	-	Yes	Almost-empty flag, set threshold to 0
AlmostEmptyTh	[RASIZE-1:0]	I	-	Yes	Almost-empty flag, set threshold to 1
AlmostFullSetTh	[ASIZE-1:0]	I	-	Yes	Almost-full flag, set threshold to 1
AlmostFullClrTh	[ASIZE-1:0]	I	-	Yes	Almost-full flag, set threshold to 0
AlmostFullTh	[ASIZE-1:0]	I	-	Yes	Almost-full flag, set threshold to 1
Q	[RDSIZE-1:0]	Output	-	No	Read Data

Signal	Data Width	I/O	Initial State	Optional	Description
Empty	1	Output	1	No	Empty flag
Full	1	Output	0	No	Full flag
Wnum	[ASIZE: 0]	Output	0	Yes	Write data number
Rnum	[RASIZE: 0]	Output	0	Yes	Readable data number
Almost_Empty	1	Output	1	Yes	Almost-empty flag
Almost_Full	1	Output	0	Yes	Almost-full flag
ERROR	2	Output	0	Yes	ECC check output

3.2 FIFO SC HS IP Port

A description of the FIFO SC HS IP I/O is presented in Table 3-2.

Table 3-2 FIFO SC HS IP IO Post

Signal	Data Width	I/O	Initial State	Optional	Description
Data	[DSIZE-1:0]	Input	-	No	Input data
Clk	1	Input	-	No	Write clock
WrEn	1	Input	-	No	Write enable
RdEn	1	Input	-	No	Read enable
Reset	1	Input	-	Yes	Reset, active high
AlmostEmptySetTh	[ASIZE-1:0]	Input	-	Yes	Almost-empty flag, set threshold to 1
AlmostEmptyClrTh	[ASIZE-1:0]	Input	-	Yes	Almost-empty flag, set threshold to 0
AlmostEmptyTh	[ASIZE-1:0]	Input	-	Yes	Almost-empty flag, set threshold to 1
AlmostFullSetTh	[ASIZE-1:0]	Input	-	Yes	Almost-full flag, set threshold to 1
AlmostFullClrTh	[ASIZE-1:0]	Input	-	Yes	Almost-full flag, set threshold to 0
AlmostFullTh	[ASIZE-1:0]	Input	-	Yes	Almost-full flag, set threshold to 1
Q	[DSIZE-1:0]	Output	-	No	Read Data
Empty	1	Output	1	No	Empty flag
Full	1	I	0	No	Full flag
Wnum	[ASIZE: 0]	I	0	Yes	Write data number
Almost_Empty	1	Output	1	Yes	Almost-empty flag

Signal	Data Width	I/O	Initial State	Optional	Description
Almost_Full	1	Output	0	Yes	Almost-full flag
ERROR	2	Output	0	Yes	ECC check output

4 Timing Description

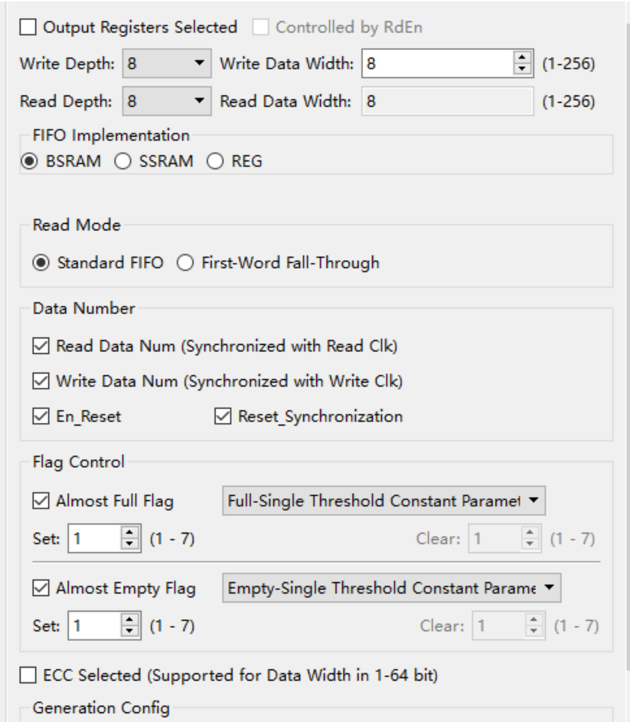
This chapter describes FIFO HS IP and FIFO SC HS IP timing during read and write operations.

In practical applications, users may configure different output control signals according to their own requirements. The Read/Write control occasions are different. The most common and important signals including, input, output, empty & full, almost-empty, almost-full are described in this chapter.

4.1 FIFO HS IP Signal Timing

Figure 4-2 shows a read/write operation of the FIFO HS in the configuration shown in Figure 4-1.

Figure 4-1 FIFO HS Configuration



As shown in Figure 4-2, when the FIFO is not full, the write enable is pulled up and the data is written to FIFO. The FIFO is full after eight writes and the write enable will be masked. The Full signal is pulled up, and data cannot be written at this time. When the FIFO is not empty, the read enable is pulled up and the data written to the FIFO is read to Q in turn. When the eighth number is read, the read enable will be masked. The Empty signal is pulled up, and the data cannot be read at this time.

Figure 4-2 FIFO HS IP Timing

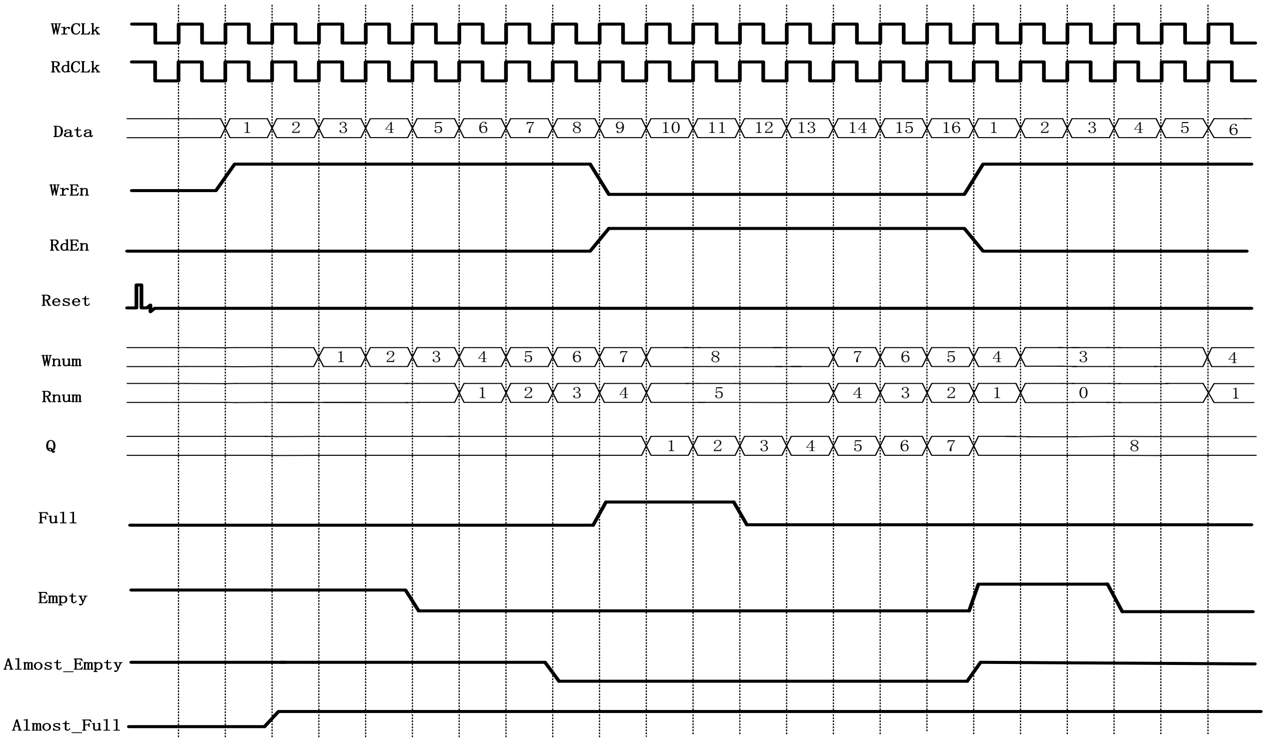
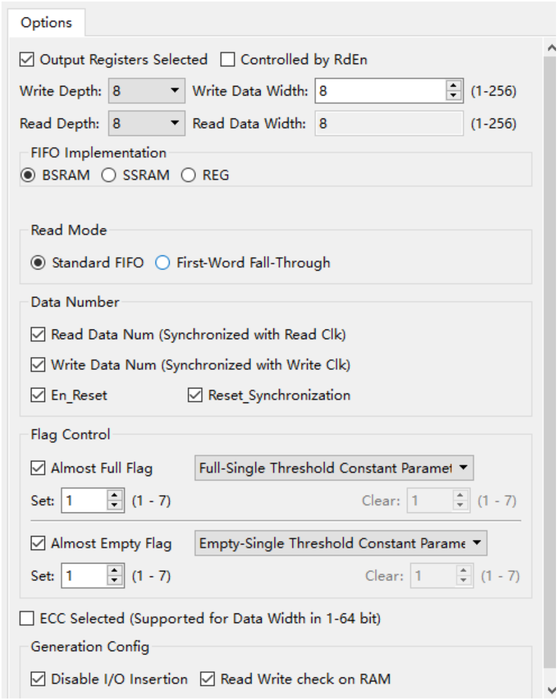


Figure 4-4 shows a read/write operation of the FIFO HS in the configuration shown in Figure 4-3.

Figure 4-3 FIFO HS Configuration



As shown in Figure 4-3, the output register is selected, so the read data is one cycle later than when the output register is not configured, and the last data will be output.

Figure 4-4 FIFO HS IP Timing

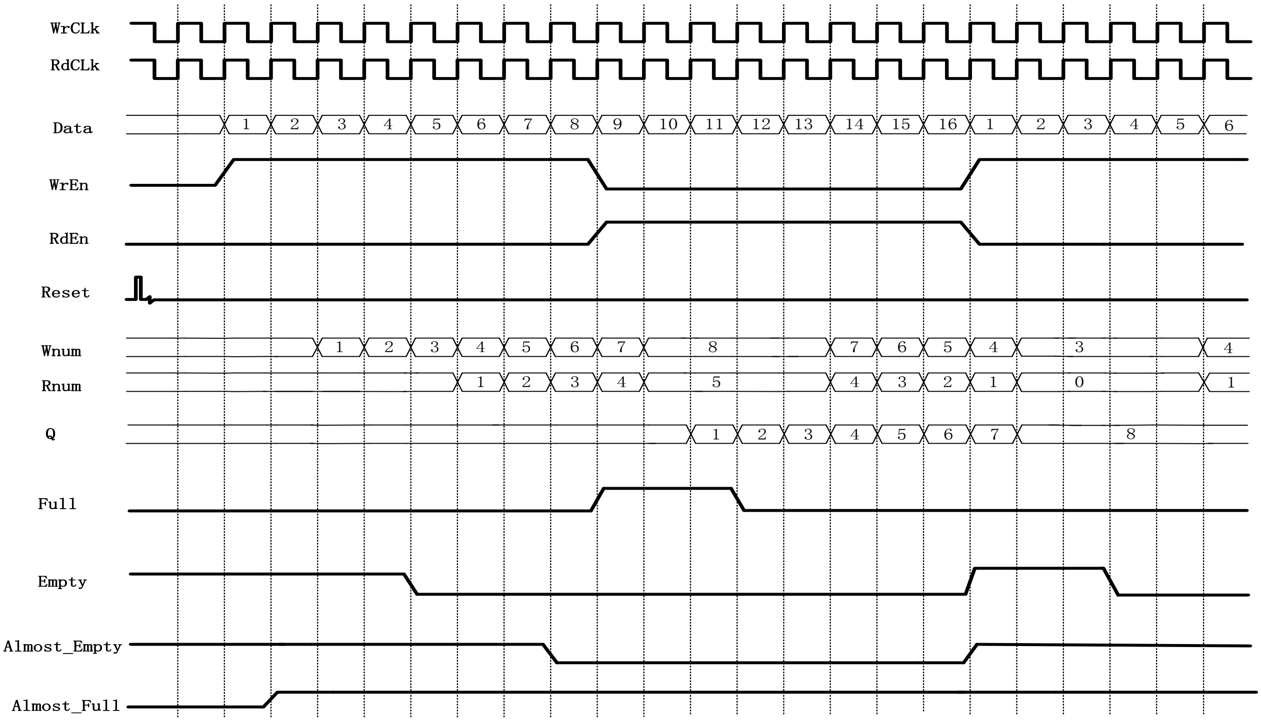
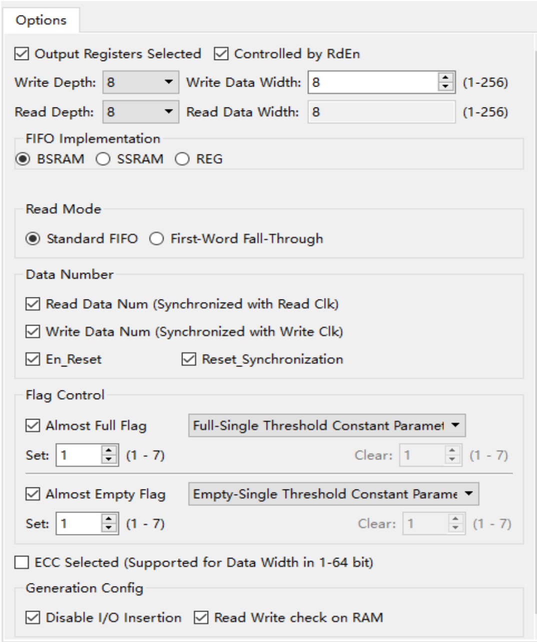


Figure 4-6 shows a read/write operation of the FIFO HS in the configuration shown in Figure 4-5.

Figure 4-5 FIFO HS Configuration



As shown in Figure 4-5, the output register is selected and the read enable control is also selected, i.e., the output register is controlled by RdEn, so the read data is one cycle later than when the output register is not configured, and the last data will not be output under the first read enable, and this data will be the first data output of the next read enable.

Figure 4-6 FIFO HS IP Timing

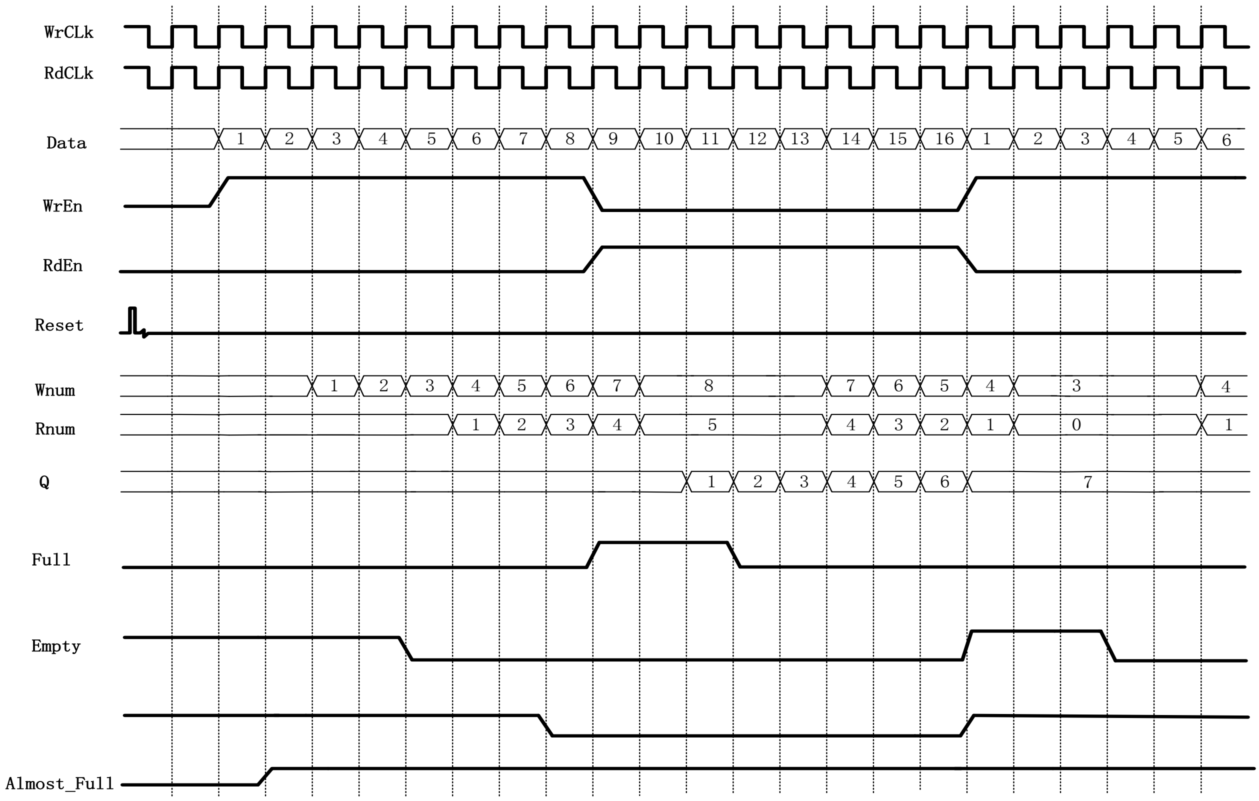
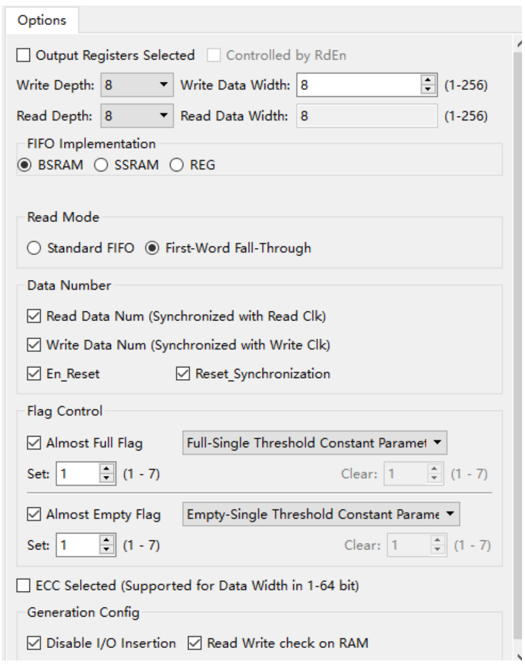


Figure 4-8 shows a read/write operation of the FIFO HS in the configuration shown in Figure 4-7.

Figure 4-7 FIFO HS Configuration



As shown in Figure 4-7, First-Word Fall-Through selected, it will put the first number written on the output data bus immediately when the FIFO is non-empty regardless of whether there is a read enable signal or not, and it will output the other data written in order when the read enable is pulled up.

Figure 4-8 FIFO HS IP Timing

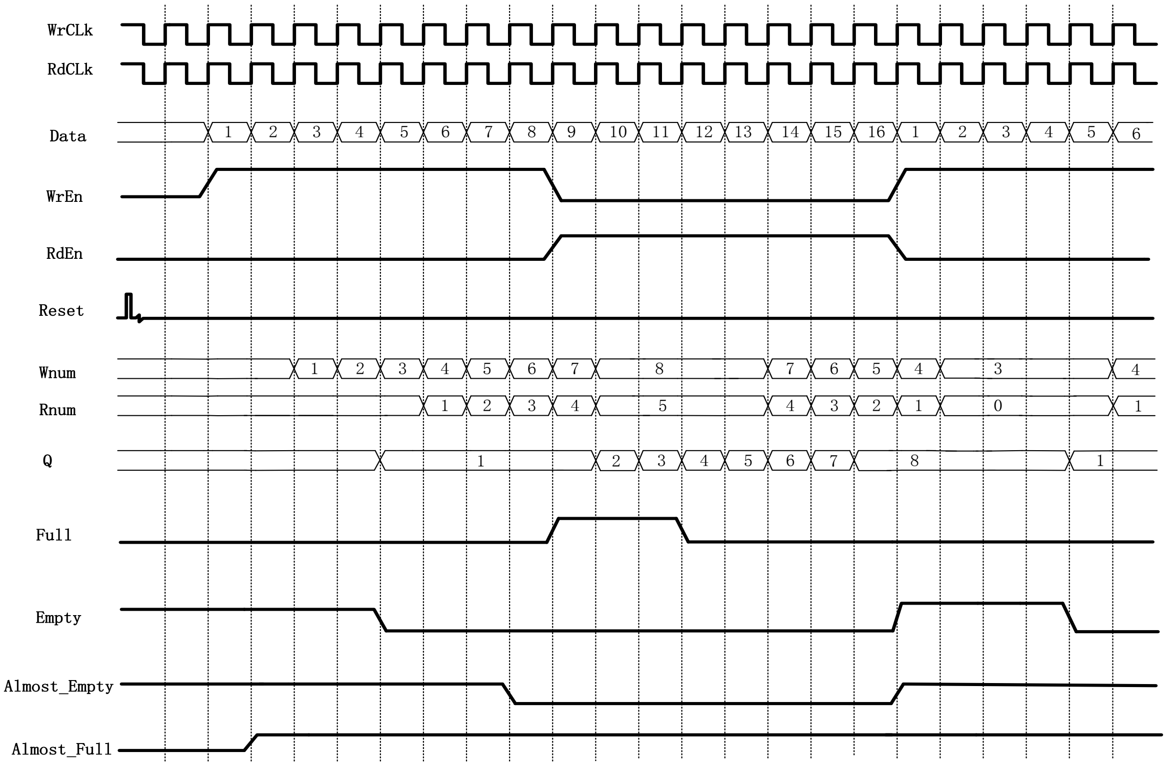
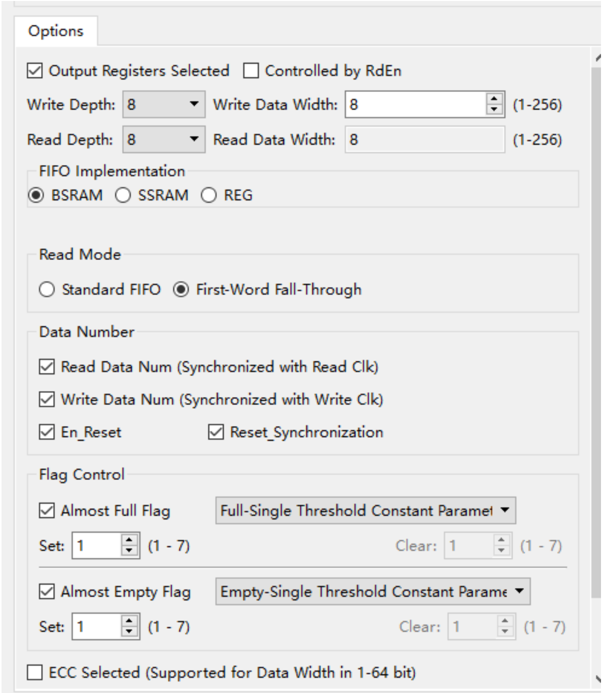


Figure 4-9 shows a read/write operation of the FIFO HS in the configuration shown in Figure 4-8.

Figure 4-9 FIFO HS Configuration



As shown in Figure 4-8, the FWFT and output register are selected, which means that the output data of FIFO in FWFT mode is registered at one level before output.

Figure 4-10 FIFO HS IP Timing

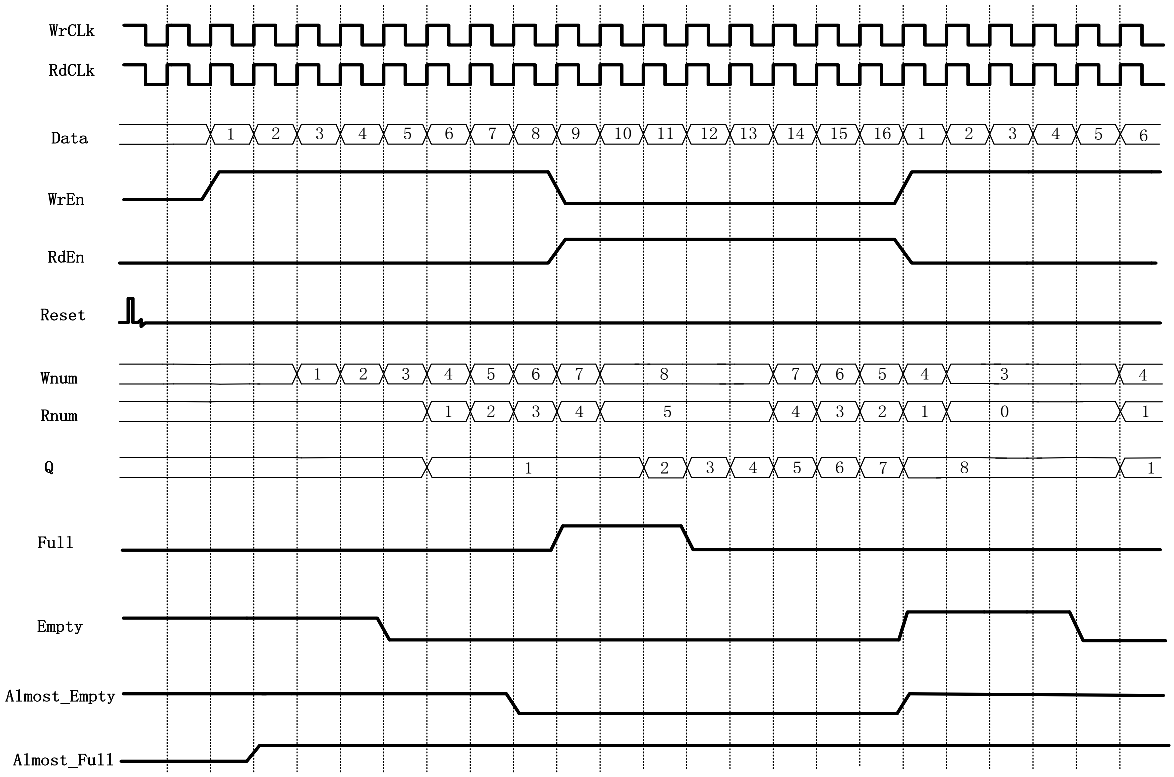
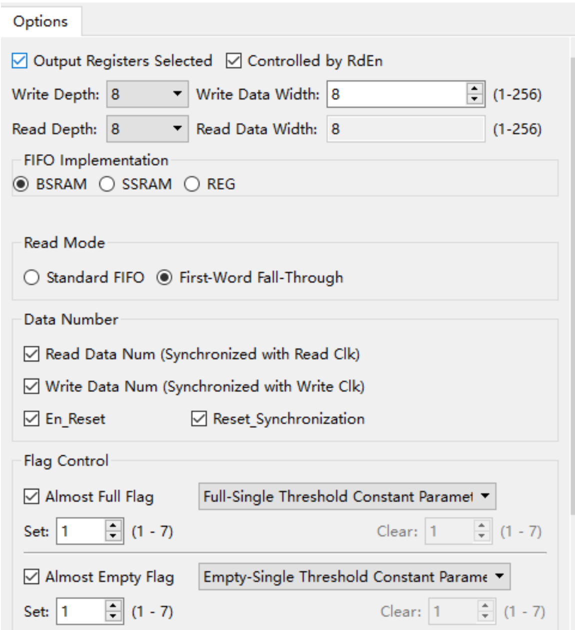


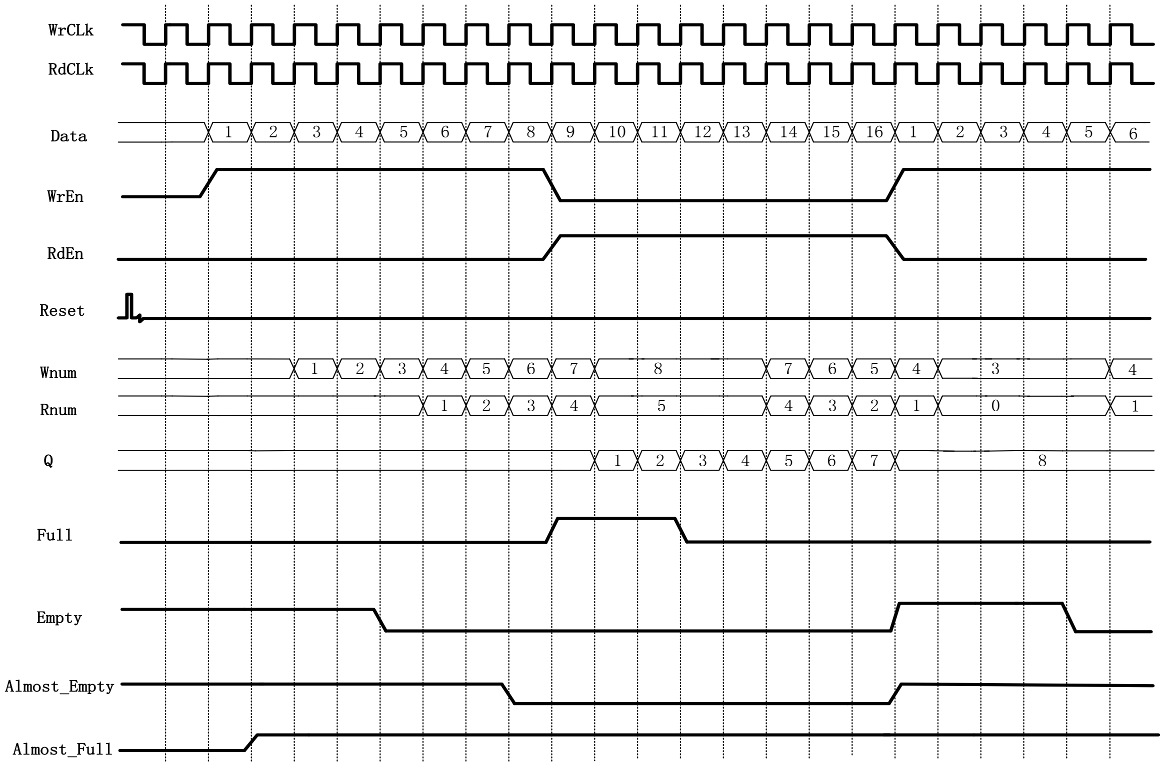
Figure 4-12 shows a read/write operation of the FIFO HS in the configuration shown in Figure 4-11.

Figure 4-11 FIFO HS Configuration



As shown in Figure 4-11, the FWFT mode, the output register and read enable control are selected, i.e., the output data of FIFO in FWFT mode is registered at one level, but the registered data is not output immediately, but output under the control of read enable.

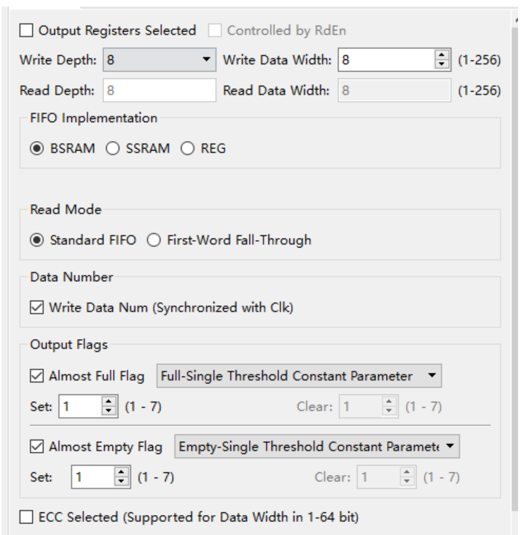
Figure 4-12 FIFO HS IP Timing



4.2 FIFO SC HS IP Signal Timing

Figure 4-14 shows a read/write operation of the FIFO SC HS in the configuration shown in Figure 4-13.

Figure 4-13 FIFO SC HS Configuration



As shown in Figure 4-13, when the FIFO is not full, the write enable is pulled up and the data is written to the FIFO; when the FIFO is full after eight writes, the write enable will be masked and the Full signal is pulled up, and no more data can be written at this time. The Empty signal is pulled up, and no more data can be read at this time.

Figure 4-14 FIFO SC HS IP Timing

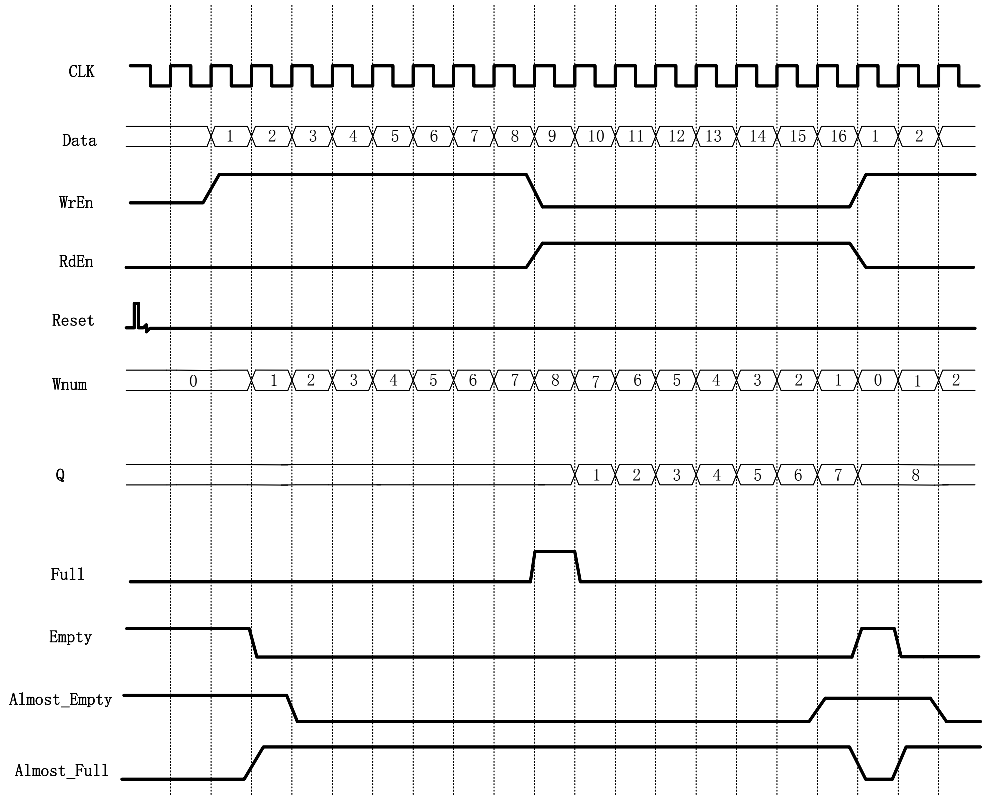
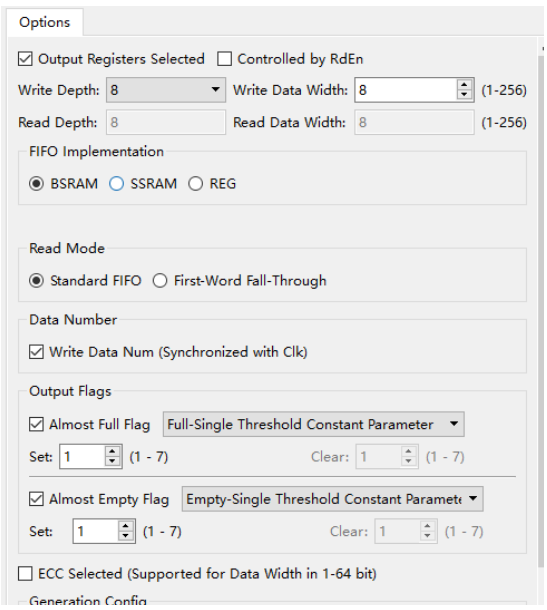


Figure 4-16 shows a read/write operation of the FIFO SC HS in the configuration shown in Figure 4-15.

Figure 4-15 FIFO SC HS Configuration



As shown in Figure 4-15, the output register is selected, so the read data is one cycle later than when the output register is not configured, and the last data will be output.

Figure 4-16 FIFO SC HS IP Timing

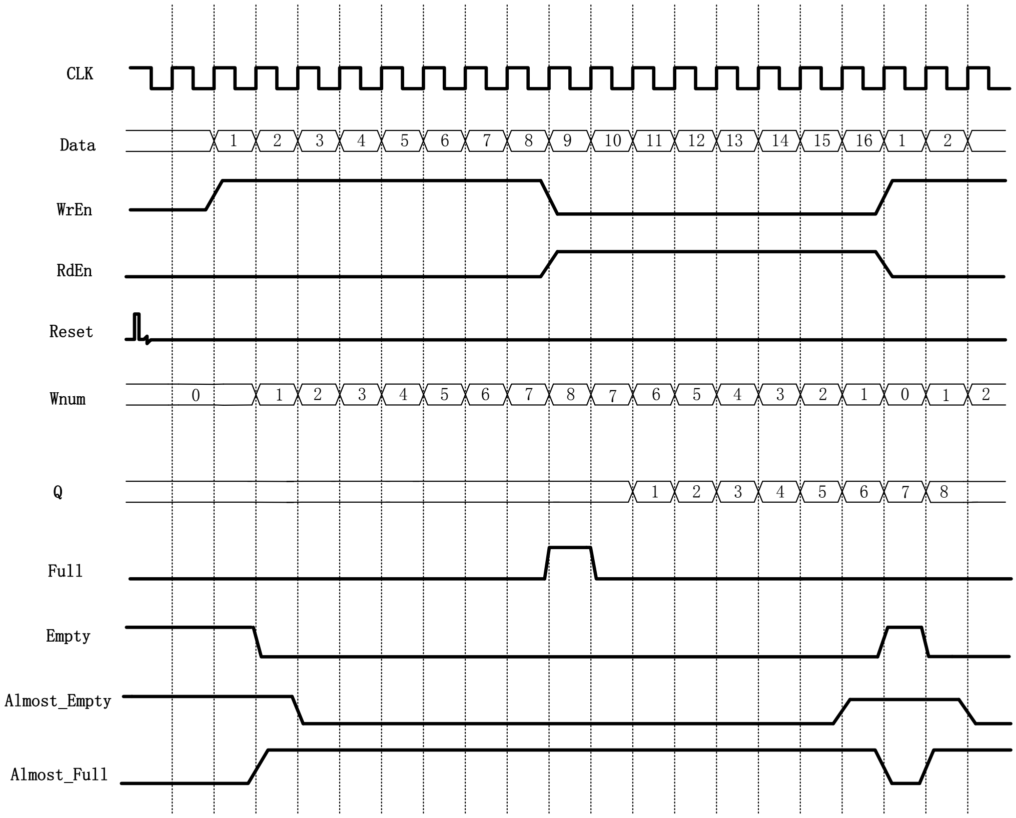
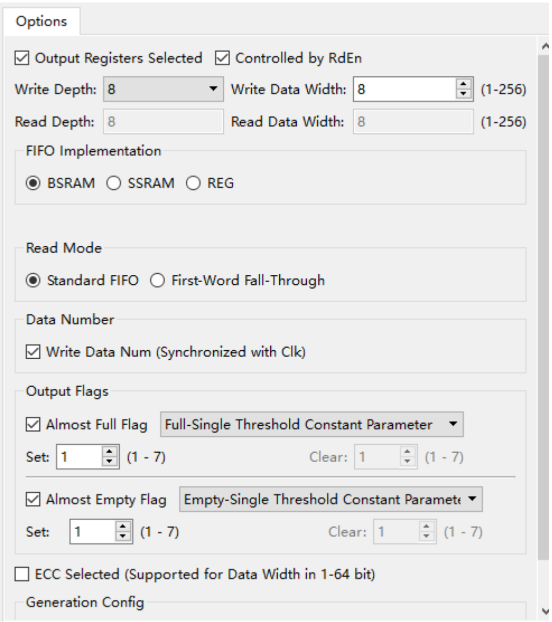


Figure 4-18 shows a read/write operation of the FIFO SC HS in the configuration shown in Figure 4-17.

Figure 4-17 FIFO SC HS Configuration



As shown in Figure 4-17, the output register and the read enable control are selected, i.e., the output register is controlled by Raden, so the read data is one cycle later than when the output register is not configured, and the last data will not be output under the first read enable, and this data will be the first data output of the next read enable.

Figure 4-18 FIFO SC HS IP Timing

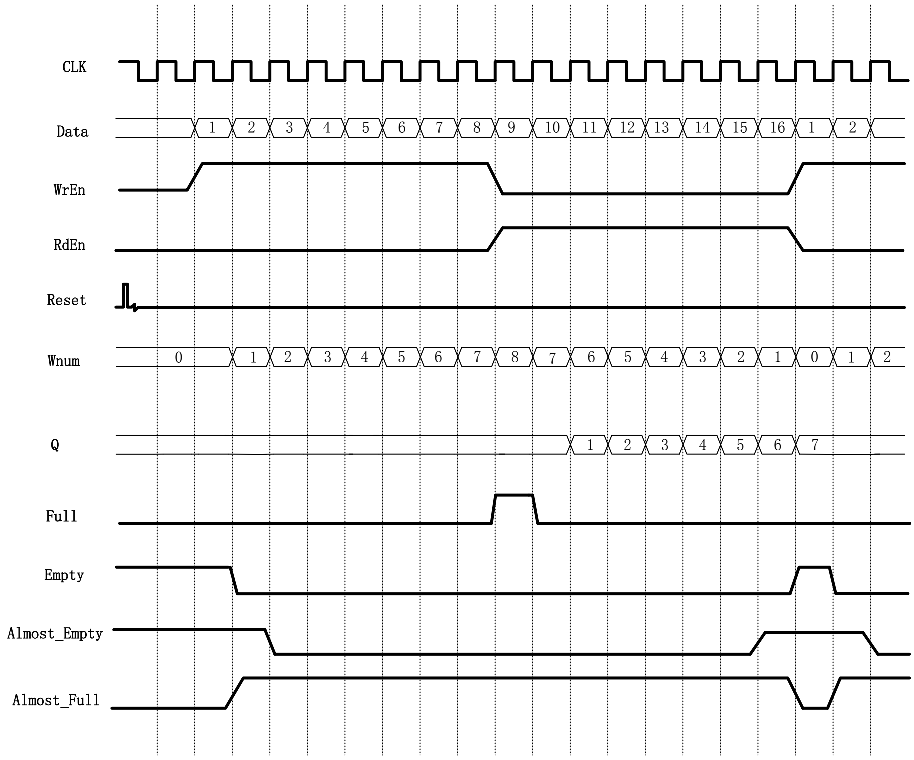
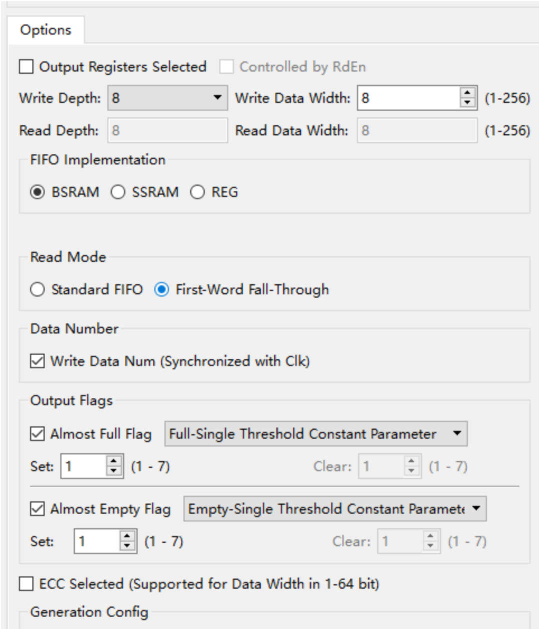


Figure 4-20 shows a read/write operation of the FIFO SC HS in the configuration shown in Figure 4-19.

Figure 4-19 FIFO SC HS Configuration



As shown in Figure 4-19, select the FWFT mode. When the FIFO is non-empty, the first data written will be immediately put on the output data bus regardless of whether there is a read enable signal or not, and the rest of the data written will be output in order when the read enable is pulled up.

Figure 4-20 FIFO SC HS IP Timing

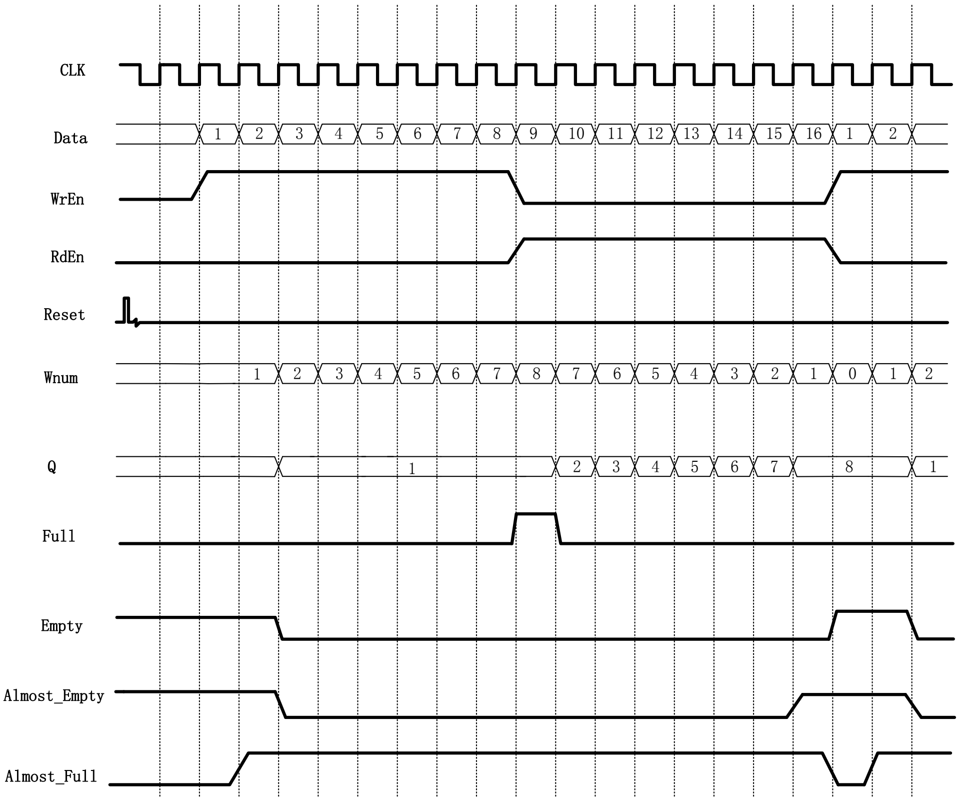
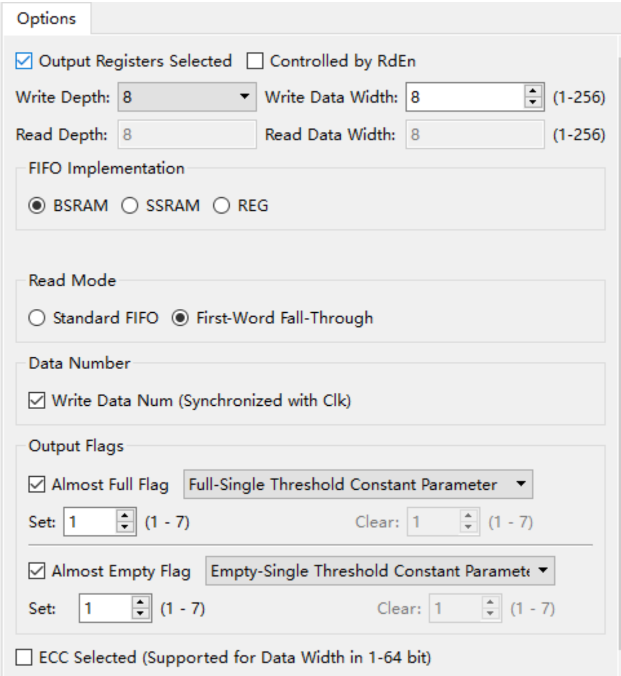


Figure 4-22 shows a read/write operation of the FIFO SC HS in the configuration shown in Figure 4-21.

Figure 4-21 FIFO SC HS Configuration



As shown in the Figure 4-21, the FWFT mode and output register are selected, i.e., the output data of FIFO in FWFT mode is registered at one level before output.

Figure 4-22 FIFO SC HS IP Timing

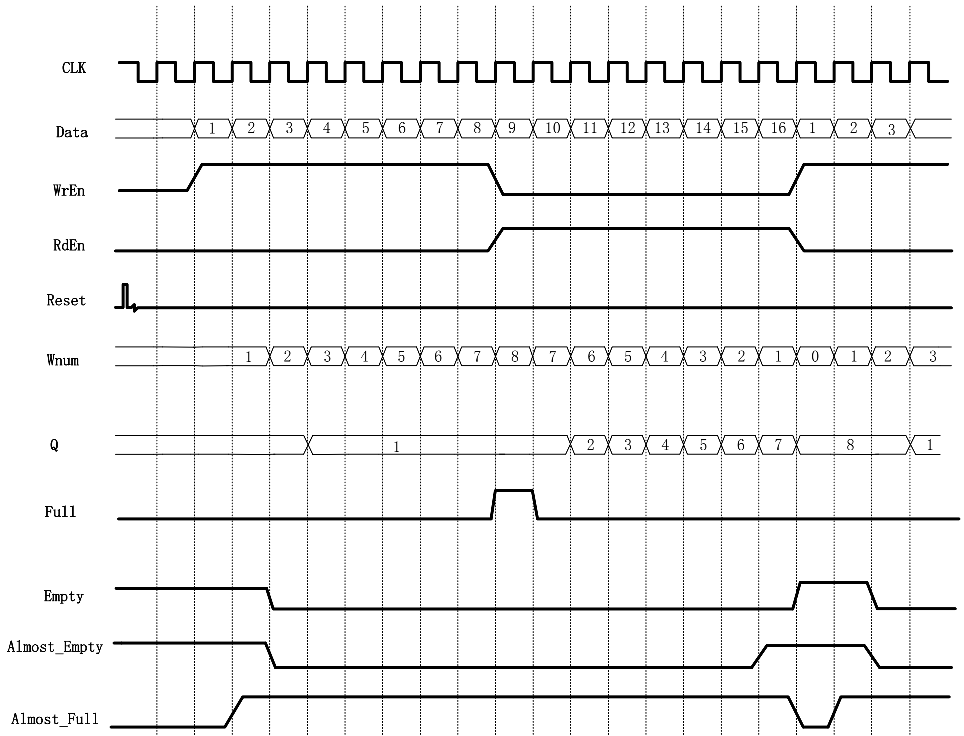
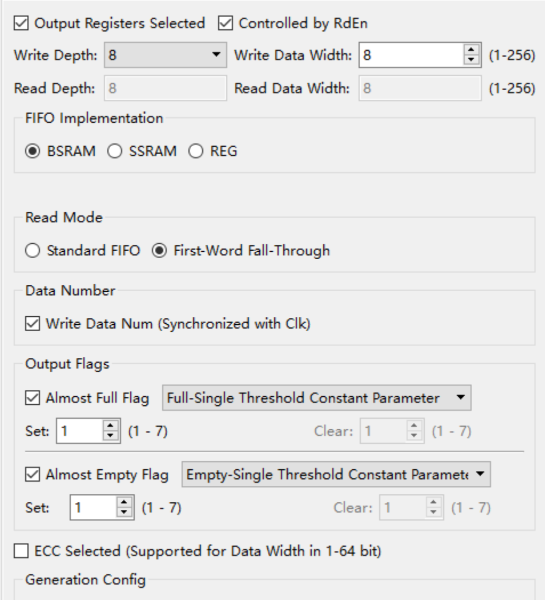


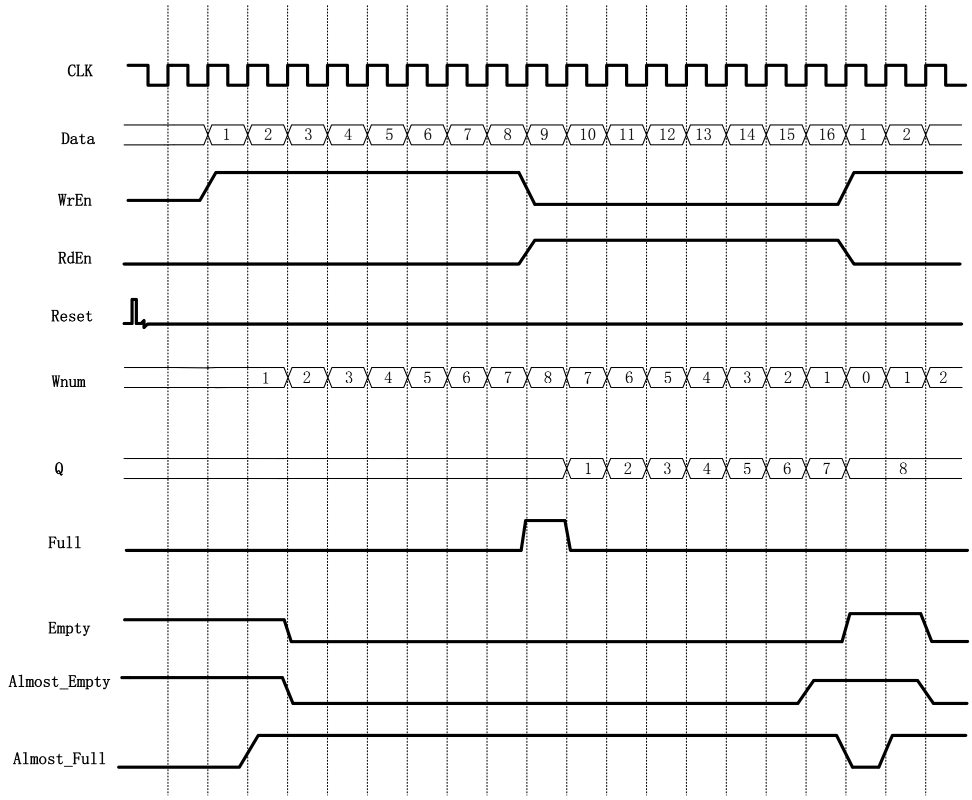
Figure 4-24 shows a read/write operation of the FIFO SC HS in the configuration shown in Figure 4-23.

Figure 4-23 FIFO SC HS Configuration



As shown in Figure 4-23, the FWFT mode, output register and read enable control are selected, that is, the output data of FIFO in FWFT mode is registered at one level. The registered data is not output immediately, but output under the control of read enable.

Figure 4-24 FIFO SC HS IP Timing



5 FIFO HS/FIFO SC HS IP Configuration

Start "IP Core Generator" from the "Tools" menu in Gowin Software; then configure and generate FIFO HS IP or FIFO SC HS IP. The FIFO HS/FIFO SC HS IP configuration interface and interface parameters are described below.

5.1 FIFO HS IP Configuration

FIFO HS IP configuration interface is as shown in Table 5-1.

Figure 5-1 FIFO HS IP Configuration Interface

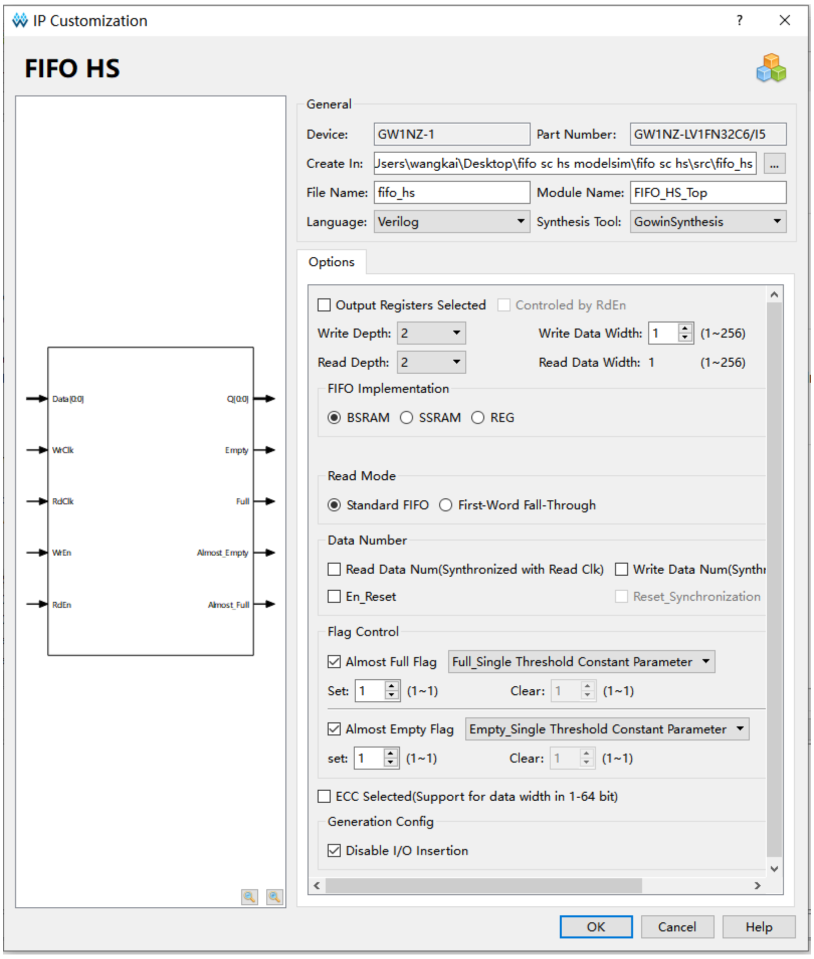


Table 5-1 FIFO HS IP Configuration Parameter

Options	Name		Description	Remarks
File	Target Device		-	-
	File Name		FIFO HS files name	-
	Module Name		FIFO HS top module name	-
	Create In		-	When a FIFO HS IP model is created, you can choose whether to add it to the current project.
Options	FIFO Implementation	Block SRAM	Select to use Block SRAM, Shadow SRAM, or LUT	-
		Shadow SRAM		
		LUT		
	Write Depth		Write Data Depth	When asynchronous FIFO HS IP write data bit-width is different from the read data bit-width, read data bit-Width will be calculated automatically using the following formula: Write Depth x Write Data Width / Read Depth.
	Write Data Width		Write Data Width	
	Read Depth		Read Data Depth	
	Read Data Width		Read Data Width	
	Read Mode	Standard FIFO	Standard FIFO	Standard FIFO reads and writes timing in accordance with the timing standard. FWFT FIFO will write the first number immediately on the output data
		First-Word Fall-Through	FWFT FIFO	

Options	Name			Description	Remarks
					bus regardless of whether there is a read enable signal. Read enable pulled high will be written in order to output the other data.
	Data Num	Read Data Num		Write data number	Add Wnum output when it is valid.
		Write Data Num		Read data number	Add Rnum output when it is valid.
		En_Reset		Enable reset	When enable, read/write reset respectively, add WrReset and RdReset input.
		Reset_Synchronization		Same reset	When En_Reset is valid, Reset_Synchronization is optional; If selected, it indicates that a reset is used to add the input Reset.
	Flag Control	Almost Full Flag		Almost-full flag enable	Add Almost_Full output when it is valid.
		Almost Full Flag	Full-Single Threshold Constant Parameter	When static almost-full single threshold is valid, Set is valid.	When Almost Full Flag is valid, users can select one out of four: Full-Single Threshold Constant Parameter,
			Full-Dual Threshold Constant Parameters	When static almost-full dual threshold is valid, Set and Clear are valid.	

<table>
<tr><th>Options</th><th colspan="3">Name</th><th>Description</th><th>Remarks</th></tr>
<tr><td rowspan="9"></td><td rowspan="9"></td><td rowspan="2"></td><td>Full-Single Threshold Input Parameter</td><td>When Dynamic almost-full single threshold input is valid, add AlmostFullTh input.</td><td rowspan="2">Full-Dual Threshold Constant Parameters, Full-Single Threshold Input Parameter, and Full-Dual Threshold Input Parameters.</td></tr>
<tr><td>Full-Dual Threshold Input Parameters</td><td>When Dynamic almost-full dual threshold input is valid, add AlmostFullSetTh and AlmostFullClrTh input.</td></tr>
<tr><td colspan="2">Set</td><td>Set threshold to 1 when almost-full</td><td rowspan="2">Select single constant threshold, Set is valid
Select dual constant threshold, Set and Clear are valid.</td></tr>
<tr><td colspan="2">Clear</td><td>Set threshold to 0 when almost-full</td></tr>
<tr><td colspan="2">Almost Empty Flag</td><td>Almost-empty flag enable</td><td>Add Almost_Empty output when it is valid.</td></tr>
<tr><td rowspan="4">Almost Empty Flag</td><td>Empty-Single Threshold Constant Parameter</td><td>When static almost-empty single threshold is valid, Set is valid.</td><td rowspan="4"></td></tr>
<tr><td>Empty-Dual Threshold Constant Parameters</td><td>When static almost-empty dual threshold is valid, Set and Clear are valid.</td></tr>
<tr><td>Empty-Single Threshold Input Parameter</td><td>When Dynamic almost-empty single threshold input is valid, add AlmostEmptyTh input.</td></tr>
<tr><td>Empty-Dual Threshold Input Parameter</td><td>When Dynamic almost-empty dual threshold input is valid, add AlmostEmptySetTh and AlmostEmptyClrTh input.</td></tr>
</table>

Options	Name		Description	Remarks
		Set	Set threshold to 1 when almost-empty	Select single constant threshold, Set is valid; Select dual constant threshold, Set and Clear are valid.
		Clear	Set threshold to 0 when almost-empty	
	ECC Selected		ECC	Valid when data width is 1-64bit; add ERROR output.
	Output Register Selected		Output register can be selected.	When it is valid, data output will be a cycle later. BSRAM output needs to meet the delay from clock to output. When Output Register enable is on, the delay will be met automatically; when Output Register enable is off, the delay from clock to output needs to be met to avoid sampling metastable data.
	Controlled by RdEn		Output is controlled by RdEn.	When Output Register Selected is valid, it can be selected.

Note!

- To enable the ECC function, data bit width needs to be less than or equal to 64bit;
- Different output control signals and thresholds can be configured according to the requirements. Table 5-1 shows the FIFO IP configuration.
- The Read Write check on RAM function is to prevent read and write conflicts in RAM, see FPGA or Synplify Pro user guide for details.

5.2 FIFO SC HS IP Configuration

Figure 5-2 shows the FIFO SC HS IP configuration interface.

Figure 5-2 FIFO SC HS IP Configuration Interface

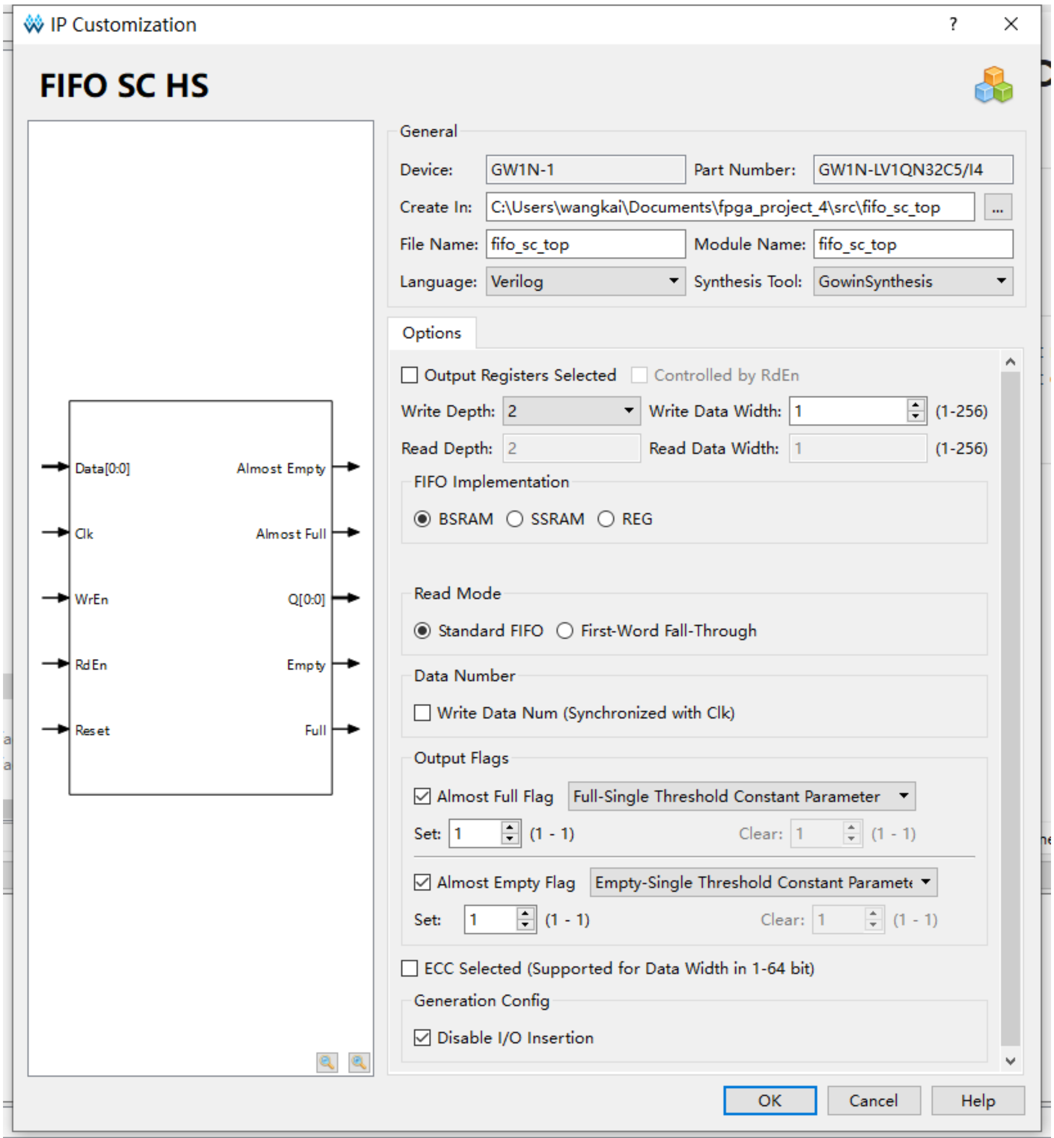


Table 5-2 FIFO SC HS IP Configuration Parameter

Options	Parameter		Description	Notes
File	Target Device		-	-
	Module Name		FIFO SC HS top module name	-
	File Name		FIFO SC HS files name	-
	Create In		-	When a FIFO SC HS IP model is created, you can choose whether to add it to the current project.
Options	FIFO Implementation	Block SRAM	Storage structure is implemented by Block SRAM.	-
		Shadow SRAM	Storage structure is implemented by Shadow RAM.	-
		LUT	Storage structure is implemented by registers.	-
	Write Depth		Write Data Depth	FIFO SC HS IP requires that write data bit-width and depth is equal to read data bit-width and depth. There’s no need to code Read Depth and Read Data Width, because these are equal to Write Depth and Write Data Width.
	Write Data Width		Write Data Width	
	Read Depth		Read Data Depth	
	Read Data Width		Read Data Width	
	Read Mode	Standard FIFO	Standard FIFO	Standard FIFO reads and writes timing in accordance with the timing standard. FWFT

<table>
<tr><th>Options</th><th colspan="3">Parameter</th><th>Description</th><th>Notes</th></tr>
<tr><td rowspan="9"></td><td></td><td colspan="2">First-Word Fall-Through</td><td>FWFT FIFO</td><td>FIFO will write the first number immediately on the output data bus regardless of whether there is a read enable signal. Read enable pulled high will be written in order to output the other data.</td></tr>
<tr><td>Data Num</td><td colspan="2">Write Data Num (Synchronized with Clk)</td><td>Write data number</td><td>Add Rnum output when it is valid.</td></tr>
<tr><td rowspan="4">Output Flags</td><td rowspan="4">Almost Full Flag</td><td>Full-Single Threshold Constant Parameter</td><td>When static almost-full single threshold is valid, Set is valid.</td><td rowspan="4">When Almost Full Flag is valid, users can select one out of four: Full-Single Threshold Constant Parameter, Full-Dual Threshold Constant Parameters, Full-Single Threshold Input Parameter, and Full-Dual Threshold Input Parameters.</td></tr>
<tr><td>Full-Dual Threshold Constant Parameters</td><td>When static almost-full dual threshold is valid, Set and Clear are valid.</td></tr>
<tr><td>Full-Single Threshold Input Parameter</td><td>When Dynamic almost-full single threshold input is valid, add AlmostFullTh input.</td></tr>
<tr><td>Full-Dual Threshold Input Parameters</td><td>When Dynamic almost-full dual threshold input is valid, add AlmostFullSetTh and AlmostFullClrTh input.</td></tr>
<tr><td></td><td colspan="2">Set</td><td>Set threshold to 1 when almost-full</td><td rowspan="2">Select single constant threshold, Set is valid; Select dual constant threshold, Set and Clear are valid.</td></tr>
<tr><td></td><td colspan="2">Clear</td><td>Set threshold to 0 when almost-full</td></tr>
<tr><td></td><td>Almost Empty Flag</td><td>Empty-Single Threshold Constant Parameter</td><td>When static almost-empty single threshold is valid, Set is valid.</td><td>When Almost Empty Flag is valid, users can select one of four options: Empty -Single</td></tr>
</table>

Options	Parameter			Description	Notes
			Empty-Dual Threshold Constant Parameters	When static almost-empty dual threshold is valid, Set and Clear are valid.	Threshold Constant Parameter, Empty-Dual Threshold Constant Parameters, Empty-Single Threshold Input Parameter, and Empty-Dual Threshold Input Parameters.
			Empty-Single Threshold Input Parameter	When Dynamic almost-empty single threshold input is valid, add AlmostEmptyTh input.	
			Empty-Dual Threshold Input Parameter	When Dynamic almost-empty dual threshold input is valid, add AlmostEmptySetTh and AlmostEmptyClrTh input.	
		Set		Set threshold to 1 when almost-empty	Select single constant threshold, Set is valid;
		Clear		Set threshold to 0 when almost-empty	Select dual constant threshold, Set and Clear are valid.
	ECC Selected			ECC	Add ERROR output when it is valid.
	Output Register Selected			Output register can be selected.	When it is valid, data output will be a cycle later. BSRAM output needs to meet the delay from clock to output. When Output Register enable is on, the delay will be met automatically; when Output Register enable is off, the delay from clock to output must to be met to avoid sampling metastable data.

Options	Parameter	Description	Notes
	Controlled by RdEn	Output is controlled by RdEn.	When Output Register Selected is valid, it can be selected.

Note!

- To enable the ECC function, data bit width needs to be less than or equal to 64bit;
- Different output control signals and thresholds can be configured according to the requirements. Table 5-2 shows the FIFO IP configuration.

6 Reference Design

The function of FIFO HS/FIFO SC HS is identical to FIFO/FIFO SC, and you can refer to the reference design of FIFO and FIFO_SC.

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